



# And in AI We Trust? A Qualitative Analysis of Trust-Building in LLM-Based Recommender Systems for Higher Education

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**Abstract.** Trust is a fundamental prerequisite for the adoption of Large Language Model (LLM)-based systems, especially in areas where knowledge is highly valued, such as higher education. While previous studies have often framed trust in conversational Artificial Intelligence (AI) as an extension of interpersonal trust, emphasizing anthropomorphism and social bonding, this study challenges that assumption. Using a Design Science Research approach, we conducted a systematic literature review and qualitative expert interviews in the context of IdeaLize, for academic collaboration. Our findings demonstrate that trust in academic recommender systems is not primarily driven by human-like interaction, but rather by transparency, source traceability and the system's capacity to acknowledge its limitations. We have derived a comprehensive catalogue of requirements structured around three clusters: (1) transparency and source traceability; (2) adaptive conversational design and controlled anthropomorphizing; and (3) personalization and neutrality. Our results propose a model of situational trust calibration, where trust stems from verifiability and user control rather than affective social cues. This work redefines trust in LLM-based systems as a design issue and offers practical guidance for the responsible adoption of AI in higher education.

**Keywords:** Trust Calibration · Large Language Models · Anthropomorphism

## 1 Introduction

Imagine sitting on a plane on the runway, waiting to take off. At this moment, passengers consciously yet involuntarily choose to put their lives in the hands of a socio-technical system comprising the aircraft, the pilots, the procedures and the regulations, despite the risks and the limited control they have. This scenario encapsulates the essence of trust: the deliberate acceptance of vulnerability [1]. Trust is therefore a fundamental mechanism for reducing complexity in uncertain environments [2]. However, this foundation is fragile; misinformation can swiftly cause rejection in interpersonal relationships, and the same is true in high-stakes human-machine contexts. A notable example of this is the Boeing 737 MAX crisis, where prioritizing financial goals over safety protocols resulted in two catastrophic accidents. Beyond the tragic loss of life, these events have

damaged public trust in automated flight systems and corporate accountability [3]. Crucially, passengers do not place social trust in the aircraft itself. Rather, trust emerges from layered verification mechanisms, institutional accountability, and the constant possibility of human intervention. This raises a fundamental question for Large-Language-Model (LLM)-based systems: why is trust in conversational AI often framed as an interpersonal phenomenon rather than as a result of system-level design and evaluation? As we transition to autonomous intelligence, this challenge becomes apparent in human-computer interaction (HCI). The hallucination phenomenon in large language models—the generation of plausible yet factually inaccurate responses—poses significant risks [4, 5]. Consequently, the field has largely drawn on interpersonal trust models—benevolence, competence, and integrity—to interpret and design LLM interactions [6]. In order to promote responsible human-AI interactions, it is essential to identify the factors that influence trust. Previous studies have shown that anthropomorphism, the attribution of human traits to non-human entities, can increase trust in voice assistants and LLMs [7]. However, other scholars argue that, in order to prevent overtrust and deception, future studies should explore the effect of responses that do not anthropomorphize [7]. Although critical design requirements such as controllability, transparency and reliability have been identified [8], the relationship between these competing design principles in specialized contexts remains unclear. This conceptual gap is particularly pronounced in academia. The IdeaLize platform is a project portfolio platform that showcases academic and research projects, as well as the profiles and interests of community members. Participants interact with an AI-based recommender system on the platform, making their perspectives particularly relevant for investigating trust and AI integration. In this context, the requirements for an LLM-based recommender system differ, reflecting the specific institutional goals and decision-making processes to which the system is applied. While many quantitative studies measure trust, few qualitative studies examine these underlying processes in detail [9, 10]. This study analyses the requirements for an LLM-based recommender system within the IdeaLize platform, which supports students, academic staff and professors by recommending relevant projects and research topics in higher education. Based on qualitative expert interviews, the study identifies three key factors in the design of trustworthy LLM-based recommender systems in academic contexts. The first contribution is that user trust should not be based on the linguistic fluency of LLM responses, but on transparency and verifiability. Source traceability can be used to enable human-in-the-loop verification, which is critical in academic environments where reliability and accountability are essential. Secondly, the study shows that trust in conversational features is influenced by how well the system adapts its response style to different types of user query. Stakeholders expect a non-anthropomorphic, factual tone for precise user queries, whereas a more human-like, guiding conversational style is favored for vague or exploratory ones. This suggests that anthropomorphism is not universally beneficial but must be regulated according to context. Lastly, personalization emerges as a nuanced design challenge rather than a universally desired feature. The results reveal a tension between users who prefer highly personalized recommendations and those who are privacy-conscious and value neutrality and minimal data usage. The study highlights the need for adaptive personalization controls that allow users to choose between personalized and neutral interaction modes. Together, these findings provide

actionable design guidance for trustworthy, adaptive, LLM-based recommender systems that support both user satisfaction and responsible AI adoption in higher education contexts.

To provide context for the subsequent theoretical framework, the IdeaLize project is presented as the primary use case. IdeaLize (stands for idea + realize) is a collaborative platform that connects students, researchers and staff from different institutions. It facilitates the sharing of ideas, the development of projects, and interdisciplinary cooperation through innovative digital tools. By promoting transparent knowledge transfer and cross-faculty collaboration, IdeaLize enhances academic exchange and fosters future innovations. IdeaLize addresses this issue by providing a central, cross-institutional project portfolio platform tailored to the Heilbronn region in Germany. At its core is an LLM-based recommendation system that suggests suitable cooperation partners and relevant projects to those involved, such as students or professors. Leveraging LLM-based matching and smart recommendations, the platform connects users with the right people, skills and projects, making academic initiatives transparent and accessible.

## 2 Related Work

Theoretical understanding of trust in technology is fundamentally rooted in established models of interpersonal trust. A key mechanism for establishing rapport between people is the Similarity-Attraction Paradigm, which suggests that people generally trust and sympathize with those they perceive as similar to themselves [11]. In communication, perceived similarity is often fostered through linguistic and behavioral accommodation, whereby conversational partners adapt their verbal and non-verbal cues to each other subtly. These mechanisms have been extensively studied in social psychology and communication research as an effective way of building rapport and trust [9, 12]. The concept of establishing trust through behavioral congruence has a well-documented history of transference into HCI and human-robot interaction (HRI). For example, research in HRI has shown that humanoid robots which employ active listening and mirroring of human movements encourage deeper emotional engagement and reduce negative attitudes [13]. Furthermore, non-verbal synchronization has been shown to directly influence the perception of intimacy and trust in technological agents [14]. Thus, the principle of leveraging human social cues for technological agents, be it physical synchronizations or the adaptation of tone and language is a proven strategy for building user trust. Current state-of-the-art techniques apply established interpersonal trust mechanisms to LLMs, where linguistic quality and conversational fluency are the main indicators of trustworthiness [9, 10, 15]. However, as language is a fundamental tool in high-stakes fields such as medicine, law and academia, LLMs' ability to convincingly mimic linguistic nuances introduces significant ethical concerns [16]. Consequently, users may attribute technologically induced trustworthiness to fluent yet potentially inaccurate responses, thereby increasing the likelihood of blind trust and uncritical acceptance of misinformation [17]. While much prior research has focused on transferring interpersonal trust mechanisms to artificial agents, fewer studies have critically examined whether such a transfer is desirable in domains where verification, accountability, and human oversight are central, such as medicine, law, and academia. These risks have complex design implications for LLM

responses. A critical challenge lies in structuring explanations to encourage appropriate reliance without provoking over-reliance [18]. One emerging area of research is the integration of explicit trustworthiness scores generated by the model itself, which would allow it to communicate its own level of certainty regarding an answer more transparently [19]. Alongside this, various intervention strategies, such as reliance disclaimers or highlighting uncertainty, are being investigated. Preliminary results suggest that, while such interventions can reduce over-reliance, they do not necessarily improve the quality of the user's actual decision-making [20]. Another area of discussion concerns the design of AI personality. Although anthropomorphic designs often increase user engagement [21], studies indicate that users prefer neutral, non-anthropomorphized responses when dealing with precise, technical queries [22]. Furthermore, the effectiveness of anthropomorphism appears to be highly dependent on the individual user; similarity-attraction effects cause users to prefer different interaction styles based on their own personality traits [23, 24]. Since LLMs can infer personality from text and adapt their interaction styles accordingly, new possibilities for personalized trust design emerge [25, 26]. In the field of recommender systems, LLMs enable a new generation of dialogue-based interfaces that provide recommendations and their underlying rationale [27], potentially increasing transparency [28]. However, a significant research gap remains. Although the debate surrounding general human-like versus factual strategies is well-established, the interaction between these competing design principles within the context of a language-based recommender system for an academic platform like IdeaLize remains unclear. The most relevant factors for trust in this specific application field remain unknown, as does the optimal prioritization of these factors and their translation into actionable system requirements. Specifically, research on how adaptive response strategies, such as factual-neutral styles for concrete queries versus conversational-consultative support for vague enquiries, can stabilize trust in sensitive academic environments is lacking. The present study addresses this gap by deriving targeted design requirements for a trustworthy, LLM-based recommender system based on qualitative expert interviews.

### 3 Methodology: A Design Science Research Approach

To address the research gap identified in the introduction, this study adopts the Design Science Research (DSR) approach. The DSR paradigm focuses on the creation and evaluation of innovative artefacts to solve identified problems [29]. A rigorous understanding of the problem domain is essential within this framework. To achieve this, the study draws on a systematic literature review that informed the conceptual basis of trust, transparency and reliance in LLM-based systems. The literature review findings were then used to derive theoretically informed design dimensions to guide the development of the semi-structured interview protocol. Specifically, constructs identified in previous studies, such as anthropomorphism, explainability, source transparency and appropriate reliance, were translated into exploratory interview prompts that addressed how experts perceive, evaluate and interact with LLM-based recommendation systems in academic contexts. This approach ensured that the interviews systematically examined how established theoretical concepts manifest in practice, rather than merely collecting opinions.

3.1 Systematic Literature Review

We conducted a systematic literature review (SLR) to establish the theoretical basis for the requirements of the IdeaLize platform. The SLR focused on the intersection of trust, personalization and large language models (LLMs). We followed the evidence-based guidelines outlined by Kitchenham, [30] to ensure a transparent, reproducible and rigorous search strategy. As shown in Fig. 1, the review was carried out in three main phases: planning, conducting and extracting the relevant paper. To capture the multifaceted nature of the research problem, two distinct SLRs were conducted. The first focused on trust and anthropomorphism, examining human-like perceptions and their influence on user vulnerability and acceptance. The second focused on trust in LLM-based recommender systems. This examined technical design requirements such as transparency, controllability, and reliability in recommendation contexts.



Fig. 1. Procedure used for the systematic literature review

**Search Strategy and Database Selection.** The SLR execution began with the planning phase, which centered on defining four distinct research questions (RQs), as presented in Table 1. These questions provided the thematic framework for the subsequent search strategy. Following the guidelines established by Kitchenham, [30] the search process was conducted systematically in three primary stages. First, key terms were extracted directly from the RQs. To ensure maximum coverage and minimize the risk of excluding relevant studies, these terms were supplemented with synonyms, technical abbreviations and alternative spellings. Boolean operators were then employed to structure the search logic. The ‘OR’ operator grouped synonyms within a single conceptual dimension, and ‘AND’ linked these distinct dimensions. Specialized search strings were developed for each RQ to facilitate multidimensional analysis of the literature. Following iterative testing and refinement within the selected databases, the final set of search strings was established (see Tables 2 and 3). The finalized strings were then applied to four major electronic databases: IEEE Xplore, ACM Digital Library, Emerald Insight and ScienceDirect. This selection ensures balanced coverage of technical implementations (IEEE and ACM) as well as behavioral and management-oriented perspectives (ScienceDirect and Emerald).

Following Kitchenham’s, [30] guidelines, keywords were derived from the RQs and expanded using synonyms and Boolean operators. Specific search strings were constructed for each research dimension to ensure a high level of precision. To ensure a

Table 1. Research questions for the SLR

| Research question (RQ) | Question                                                                                               |
|------------------------|--------------------------------------------------------------------------------------------------------|
| RQ1                    | Which established models and core factors of interpersonal trust are described in literature?          |
| RQ2                    | To what extent are these core interpersonal trust factors transferable to trust perceptions in LLMs?   |
| RQ3                    | How can these identified trust factors be integrated into the design of LLM-based recommender systems? |
| RQ4                    | What is the influence of user personality traits on the personalized LLM-based recommender systems?    |

Table 2. Search queries of the SLR on trust.

|                                                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Keywords used in the search queries of the SLR on trust                                                                                                                                                                        |
| Document Title:anthropomorphization) AND ("Document Title":Trust) AND ("Document Title":user perception) AND ("Document Title":personalization) OR ("All Metadata":personality trait) AND ("All Metadata":Large Language Model |
| AND                                                                                                                                                                                                                            |
| similarity attraction AND social influence trust AND reputation effect AND :active listening communication AND Human trust                                                                                                     |

Table 3. Search queries of the SLR on anthromorphization.

|                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Keywords used in the search queries of the SLR on anthropomorphization                                                                                                                                                                                              |
| ("large language model" OR LLM) AND ("recommender system" OR "recommendation system") AND ("higher education" OR university) AND ("hallucination" OR "misinformation")                                                                                              |
| AND                                                                                                                                                                                                                                                                 |
| ("large language model" OR LLM) AND ("recommender system" OR "recommendation system") AND ("higher education" OR university OR academic) AND (anthropomorphization OR "de-anthropomorphization" OR persona) AND ("user experience" OR UX OR "human-ai interaction") |

comprehensive overview, the search strategy was not limited to the term ‘large language model’ (LLM), since this specific terminology has only recently gained widespread academic dominance. Instead, the search included broader, related terms such as ‘artificial intelligence’ (AI), ‘chatbots’, and ‘conversational agents’. Tables 2 and 3 summarize the finalized search strings applied to the databases.

Finally, these strings were applied to the electronic databases IEEE Xplore, the ACM Digital Library, Emerald Insight and ScienceDirect. Figure 2 shows the PRISMA flow diagram detailing the overall search and selection process.

**Study Selection and Inclusion Criteria.** The selection process applied explicit criteria to filter the initial search results, ensuring they were highly relevant to the research questions. The inclusion and exclusion criteria focused on peer-reviewed journal articles,

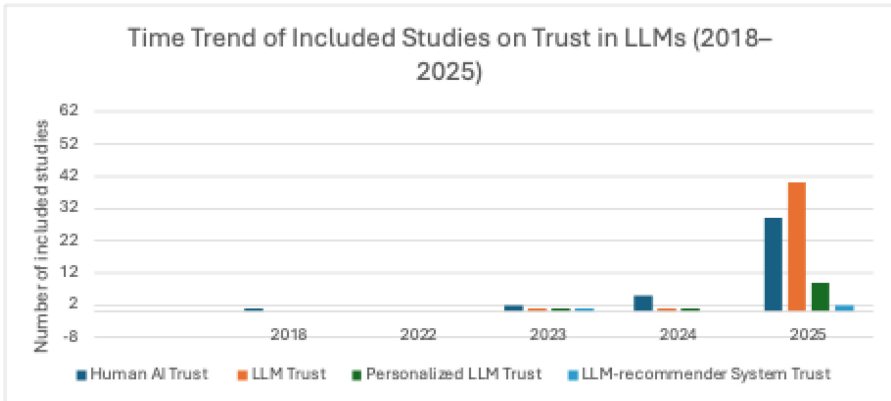
conference proceedings and book chapters. To maintain academic rigour, only papers written in English and at least four pages long were considered. A key consideration in the selection process was the concept of anthropomorphism, or the extent to which an LLM-based recommender system exhibits human-like characteristics in its communication. This is a crucial aspect for determining whether a ‘human-like’ or ‘neutral/robotic’ interface is more effective in building trust. A differentiated timeframe was applied to reflect the varying maturity of the research topics. Keywords related to interpersonal trust were assigned a 15-year timeframe, as the theoretical foundations date back to the late 1990s and early 2000s. In contrast, LLM technology and search terms related to anthropomorphism were restricted to the last five years, focusing on the rapid evolution of generative AI since the emergence of models such as ChatGPT in 2022. In total, the preliminary searches yielded 41 studies focusing on personalization, 15 studies focusing on anthropomorphism in AI interfaces and 62 studies specifically addressing trust in connection with LLMs.

**Analysis Procedure of Extracted Papers.** After selecting the relevant papers, a qualitative analysis was conducted to evaluate the quality of the studies and extract data relevant to the research questions. This procedure adhered to the final steps of Kitchenham’s (2004) systematic review protocol. Study quality was evaluated using subcategories derived directly from the research questions. Each paper was assessed using a structured scoring scheme (Yes = 1, Partially = 0.5, No = 0), which captured whether the paper addressed the following research questions:

- RQ1: Trust in human–human interaction.
- RQ2: Trust in LLM-based systems, including anthropomorphic and human-like communication strategies.
- RQ3/RQ4: Trust factors or personalization mechanisms in LLM-based recommender systems.

To contextualize the maturity of the research field, a time trend analysis was conducted. Although the literature search focused on the last five years to ensure technical relevance, publication dates were examined from 2018 onwards. The analysis revealed that research explicitly addressing trust in LLM-based systems only emerged after the introduction of transformer-based models, which justifies the focus on recent literature. Based on a qualitative assessment, the papers were classified into seven thematic categories. As there are conceptual overlaps, particularly between trust and personalization in LLM-based recommender systems, papers could be assigned to multiple categories. This categorization provides a structured overview of the field, indicating that despite extensive foundational research on trust and user traits, studies that explicitly address trust and personalization in LLM-driven recommender systems are comparatively limited. Although the literature search focused on the last five years to ensure technical relevance, the time trend analysis was extended back to 2018 to establish a baseline (see Fig. 2). The analysis shows that, before 2022, research explicitly addressing trust in Large Language Models was virtually absent, followed by a sharp increase from 2023 onwards. This pattern reflects the emergence of transformer-based architectures and publicly available generative models and justifies the focus on recent literature.

After all, LLM-based recommender systems represent a highly contemporary research field. Based on a qualitative assessment, the selected papers were classified into seven thematic categories. As there are conceptual overlaps, particularly between trust and personalization in LLM-based recommender systems, the categories were not treated as mutually exclusive. Overall, the distribution indicates a strong foundation in general trust and AI research. However, studies that explicitly address trust and personalization in LLM-driven recommender systems remain limited, which highlights the emerging nature of this research area.



**Fig. 2.** Time Trend according to the 62 included Papers on trust

### 3.2 Qualitative Research Design: Expert Interviews

To complement the findings of the literature review with practical insights, a qualitative research design was implemented. Unlike quantitative approaches, qualitative methods allow non-standardized, in-depth data to be collected, enabling complex motives and perspectives to be interpreted [31]. The study was conducted in strict accordance with the Consolidated Criteria for Reporting Qualitative Research (COREQ), [32], thereby ensuring both methodological rigour and transparent reporting.

**Stakeholder Selection and the IdeaLize Ecosystem.** Participants were selected via purposive sampling to target key stakeholders within the Heilbronn Bildungscampus ecosystem. The sample included a total of 11 people, including professors, senior managers and a CEO, from five leading organizations. Two higher education institutions, one research institution, and two associated organizations. These individuals were chosen because of their important roles in the local knowledge ecosystem and their involvement with the IdeaLize app on the Heilbronn project platform. Their expertise has particular relevance for the investigation of the intersection of institutional collaboration. All participants were provided with informed consent prior to taking part and signed a data protection and privacy statement. The study complied with applicable data protection regulations, and all interview data were anonymized for analysis and reporting purposes.

**Interview Protocol.** The study used semi-structured, guideline-based interviews. This format ensures a systematic flow of information while allowing participants to provide

open-ended, detailed responses [33]. The interview guide was developed based on the theoretical frameworks of [22, 34]. The central focus of the data collection was to investigate the question of whether enhanced anthropomorphism in Generative AI (GenAI) results in enhanced trust, or whether it could potentially diminish trust in certain professional settings. Experts were contacted via email or telephone. After giving their consent, the participants received the interview guide to help them prepare. A pre-test was conducted to minimize errors. The interviews were conducted online, digitally recorded, and subsequently transcribed using MAXQDA software.

**Data Analysis: Qualitative Content Analysis and COREQ Compliance.** The analysis was conducted using the systematic framework of Qualitative Content Analysis, as outlined by Mayring [31]. To ensure compliance with the COREQ guidelines, the analysis was conducted using a transparent, step-by-step process. An inductive approach was chosen for category development, enabling themes to emerge directly from the material [31]. The summarizing technique was applied to reduce the transcripts to their essential core statements without losing semantic depth. The coding unit was defined as the smallest meaningful segment, while the context unit encompassed the entire utterance in order to preserve the expert's original intent. In line with COREQ requirements, the coding system (consisting of nine main codes and 25 subcodes) was constantly refined. Any modification to the categories during the analysis phase prompted a thorough re-evaluation of the material that had already been coded, in order to ensure intra-coder reliability.

## 4 Conceptualizing Trust and Personalization in LLM-Based Recommender Systems

In order to understand how LLMs can be integrated into academic environments, it is crucial to define the theoretical basis of trust and its connection with personalized interaction. This chapter provides a conceptual framework, synthesizing established trust research with the specific requirements of LLM-recommender Systems. Due to their dialogic nature, LLMs are often considered to be social actors. This has led previous studies to apply constructs of interpersonal trust to technical entities. This chapter reviews these assumptions as a conceptual baseline rather than a design prescription. The section explores how traditional trust factors, including anthropomorphism, can be linked to the requirements of an LLM-based recommender system and which other aspects become relevant in the context of human–chatbot interaction.

### 4.1 Dimensions of Trust in Human and AI Interaction

Trust has a profound impact on all social relationships and is the deciding factor in how and whether individuals interact with others [9]. In the context of academic recommender systems such as IdeaLize, trust is not a one-dimensional concept, but rather a multilevel psychological state. Building on well-established trust models, particularly the framework by [1] and its computational adaptations, trust can be categorized into three main areas: competence, benevolence and integrity [35].

**The Triangle of Trustworthiness: Competence, Benevolence and Integrity.** To assess the credibility of a trustee, whether human or AI, users look for specific psychological markers that collectively define trustworthiness. The first of these is competence, which measures a trustee's ability to fulfil the trustor's desires. It focuses primarily on the trustee's level of expertise in a specific domain [35]. In digital environments, competence is often inferred through linguistic cues, such as the quality of posts or messages. Professional cues significantly influence this perception: for example, research shows that people are far more willing to share personal or complex problems with a professional wearing domain-specific attire, such as a white coat in a medical context, than with someone in casual dress [35]. For a system like IdeaLize, this suggests that the perceived professionalism of the AI's academic tone is a vital signal of its expertise. Linked to this is the dimension of benevolence, which assesses the extent to which a trustee is motivated to act in the trustor's interests rather than following purely self-serving motives [35]. [36] argue that honesty and perceived good intentions are prerequisites for trusting an agent. In social and computational models, benevolence is reflected through responsiveness and altruistic traits, such as the willingness to engage meaningfully with the user's needs [35]. Finally, integrity is a measure of the trustee's honesty and moral consistency [35]. It involves the perception that the trustee adheres to a set of principles that the trustor deems acceptable. Furthermore, reputation plays a critical role in this triad, as individuals are more likely to trust those with an honorable reputation or a proven track record of reliable behavior [37]. For a system like IdeaLize, this suggests that the perceived professionalism of the AI's academic tone is an important indicator of its expertise. While these dimensions are established as foundational in the literature, the specific linguistic markers that convey academic benevolence in a conversational recommender remain to be investigated empirically in Sect. 5.

**Communicative Strategies Including Active Listening and Mirroring.** Trust is not static; it is built through active engagement. Active listening and mirroring are transformative communication strategies that can mitigate negative attitudes and foster intimacy [13]. In written online communication, active listening involves reflecting back the sender's impression without judgement, thereby releasing positive feelings and allowing a sense of trust to emerge [12, 38]. Although sharing deep emotions typically necessitates an existing level of trust, the mere act of listening can catalyze this development [12, 38]. Mirroring - the subtle replication of the other party's behavior, tone or gestures—reinforces emotional resonance and alignment [13]. In the context of this study, it is important to establish the point at which users perceive mirroring by a LLM-based recommender System as manipulative or disturbing. This threshold is a key component of the requirements identified in the subsequent synthesis. Applying this principle to IdeaLize raises the question of how a system can dynamically detect and adapt to user traits. This requirement is explored further in the qualitative analysis of user expectations.

**The Similarity Attraction Principle.** A key psychological driver in trust-building is the similarity-attraction effect, which states that people are naturally drawn to others they perceive as similar to themselves [11, 39]. This phenomenon is rooted in the idea that similarity reduces social uncertainty and provides a sense of cognitive consistency, making interactions feel more predictable and rewarding. When a counterpart exhibits similar characteristics, this validates the individual's worldview and social identity, thereby

fostering a sense of familiarity and comfort [11]. This effect is not limited to human-to-human interaction: research has shown that users also respond more positively to computer agents that mirror their own personality. For example, extroverted users prefer more proactive and assertive interfaces, while introverted users prefer more subtle and reserved interactions [23]. Implementing personalized communication mechanisms that address competence, benevolence and integrity can transform the system from a simple database into a trusted academic companion. However, striking the right balance between personalization and professionalism is crucial in this transition. The system must be personal enough to be engaging, yet professional enough to remain credible. The tension between personalization and neutrality is a guiding theme in the categorization of the empirical results.

## **4.2 Limits and Implications of Transferring Human Trust Models to Human–AI Interaction**

The recent introduction of highly capable LLMs is particularly noteworthy in the context of human-AI collaboration, where effective communication is paramount [40]. For IdeaLize, the challenge lies in translating these human-centric factors into a digital environment, where trust is based on perceived alignment and attentiveness [13]. This suggests that the model's ability to attentively process user input and align its tone subtly with the user's academic style is integral to conveying sincerity and building a meaningful connection. Furthermore, in the context of the IdeaLize recommender system, the similarity-attraction effect is decisive. If the LLM aligns its personality and communication style with the user's individual traits, the user is likely to rate the system as friendlier and more trustworthy and will spend more time interacting with the LLM [11]. Implementing mechanisms that address competence, benevolence and integrity through personalized communication can transform the system from a simple database into a trusted academic LLM-based recommender system.

## **4.3 The Role of Anthropomorphism**

The degree of anthropomorphism is a crucial factor in trust calibration as it creates a sense of social presence, which has been shown to increase user trust [15, 21, 24] demonstrate that simple cues such as the use of first-person pronouns convey security. [22] clarify that the impact depends on the type of interaction. Vague user queries often require high levels of anthropomorphism and social support, whereas specific, task-oriented questions require functional precision. However, [7] contend that future studies should focus on the effects of de-anthropomorphized responses to better understand their impact on user trust without human-like biases. Recent findings by [41] highlight this tension: while an anthropomorphic design does not necessarily increase perceived self-efficacy, it does lead to a significantly stronger sense of connection with the system. This suggests that humanization primarily fosters user engagement. However, for actual behavior change, such as in the field of sustainability, both anthropomorphic and factual designs show promise [41]. However, as excessive personification can trigger negative effects through social comparison [42], systems must strike a balance between emotional trust

and cognitive expertise [19]. Furthermore, preferences vary based on user personality: extroverted individuals often prefer less anthropomorphism, while those with high levels of neuroticism respond more strongly to human-like interaction partners [24]. Thus, the challenge remains to resolve the tension between social bonding and functional distance in a context-specific manner [43]. Although these models suggest that human-like communication and social alignment can foster trust, they do not address the desirability of such mechanisms in epistemically sensitive domains, such as academic decision-making. The following empirical analysis therefore examines how these assumptions apply to LLM-based recommender systems in higher education.

## **5 Results: A Requirements Catalogue for a Trustworthy LLM-Recommender System in Higher Education**

The results presented in this section are based on qualitative expert interviews conducted as part of the Design Science Research process. The interviews aimed to identify design requirements for a trustworthy LLM-based recommender system in higher education, focusing particularly on trust, transparency, anthropomorphism and personalization. A total of eleven semi-structured interviews were conducted with higher education stakeholders in the Heilbronn ecosystem. Each interview lasted between 30 and 40 min and was conducted in a one-to-one format. The interviews followed a semi-structured guide, enabling both comparability across participants and in-depth exploration of individual perspectives. The interview data were analyzed using qualitative thematic analysis. The requirement clusters presented below represent recurring patterns and design expectations that emerged across the interviews, forming the empirical basis for the proposed requirements catalogue.

### **5.1 Requirement Cluster 1: Transparency and Source Traceability**

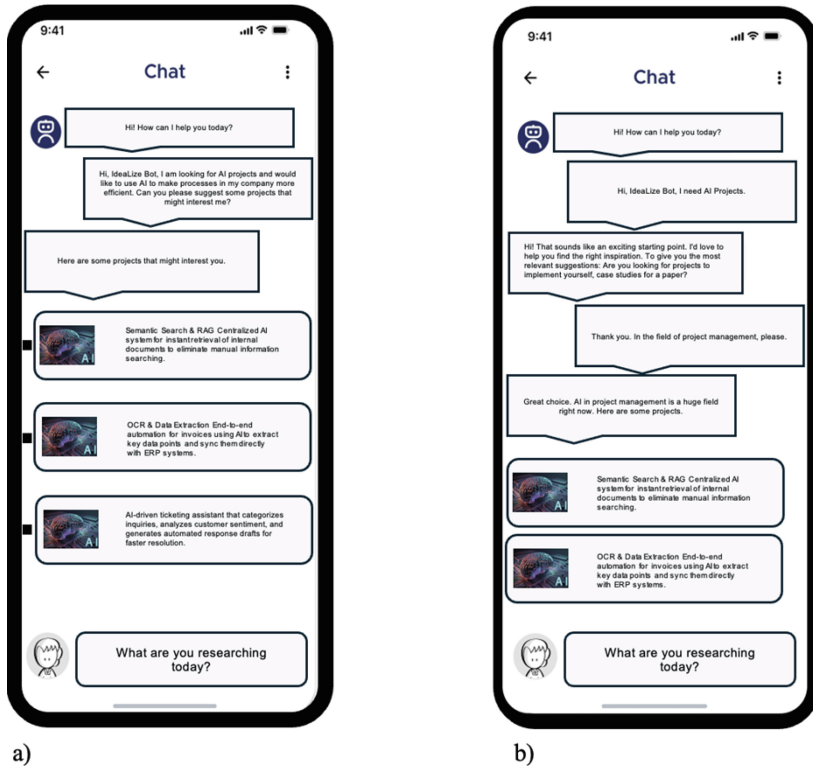
The first requirement cluster focuses on transparency and traceable sources. These emerged from user interviews as fundamental conditions for acceptance of the recommendation system. Users made it clear that their willingness to follow a recommendation is fundamentally linked to their understanding of its roots. A central aspect of this cluster is the disclosure of the database. Users need to know whether the LLM-recommender system is accessing an internal, validated university data corpus or retrieving information from the open internet. This distinction significantly affects the level of critical questioning of the results and the importance placed on the explicit citation of sources for independent follow-up research. In an academic context, interviewees consistently emphasized that an LLM-based recommender system should not only provide results, but also explain how these results were obtained, referencing specific content or sources. The majority of participants ( $n = 11$ ) articulated this expectation, highlighting transparency and traceability as fundamental prerequisites for trust in academic decision-support systems. Participants also stressed the importance of logical consistency and explainability in system communication, which correspond to the principles of explainable AI. Several interviewees ( $n = 9$ ) explicitly stated that the system should disclose which inputs, such as keywords, competency profiles or filter criteria, led to a specific

recommendation. Simple indicators, such as percentage-based match values, were considered helpful for quickly assessing relevance, provided they were accompanied by an understandable rationale. Beyond technical transparency, interviewees emphasized that recommendations must be justified in a manner that is professionally plausible. Pure similarity matching was widely regarded as insufficient; instead, participants expected explanations that aligned with their disciplinary expertise and individual project interests. Another recurring theme was the transparent communication of system limitations. The largest group of respondents ( $n = 10$ ) explicitly valued it when the LLM-based recommender system proactively acknowledged uncertainties, restricted data access or highlighted potential weaknesses, such as bias or hallucinations. Statements such as ‘I do not know’ or ‘I do not have access to this information’ were repeatedly described as trust-enhancing, as they signal epistemic humility rather than false authority.

## **5.2 Requirement Cluster 2: Adaptive Conversational Design and Anthromorphization**

The second requirement cluster focuses on adaptive conversational design. The central question is how the LLM-based recommendation system should adapt its interaction strategy to a user’s specific request. Across the interviews ( $n = 11$ ), a clear consensus emerged that static response patterns are insufficient and that adaptive interaction strategies are required to align system behavior with the specificity and context of user requests. To investigate this dynamic, the interview used a scenario in which the interviewees, acting as users in their professional roles, searched for project ideas to make processes more efficient through AI and ensure their company’s innovative capacity. This use case involved analyzing two contrasting scenarios of requirements formulation. The different design requirements are visualized in the following figure through corresponding mock-ups. In the first scenario, the LLM recommendation system is presented with a specific, fact-based search query. ‘Hello IdeaLize Bot, I am looking for AI projects and would like to use AI to make the processes in my company more efficient. Can you please suggest projects that might interest me?’ Although the query clearly formulates a goal (process efficiency), users in this case expect a structured, immediate presentation of information. Rather than engaging in a lengthy dialogue, the system should output results in a visually processed format. As shown in the corresponding mockup, users want a clear tile view for this purpose. Each card should contain essential metadata such as the project title, involved partners and best practices, to facilitate the intended exchange of experience. By contrast, the second scenario involves a vague, undefined request: ‘Hello, I need AI projects.’ According to the interviewees, providing direct results in tile form would lead to information overload in this instance, as the necessary context (e.g. industry or objectives) is missing. In this case, the LLM recommendation system must adopt a consultative approach and use adaptive dialogue design. As illustrated in the figure, the system acts as a digital mentor, exploring requirements through targeted follow-up questions. The goal is to identify with the user at which point in the process the use of AI would be most beneficial. Respondents emphasize that the LLM recommendation system must take an active lead here to lower the barrier for unclear problem definitions, only transitioning to the visual representation of project suggestions once the requirements have been clarified. In summary, adaptive conversational

design demands high flexibility: while concrete queries prioritize an efficient, tool-like presentation of results, vague queries require an anthropomorphic, directorial dialogue to guide the user towards the appropriate project idea (Fig. 3).



**Fig. 3.** A comparison of query-adaptive interaction strategies within the IdeaLize platform. (a) Direct functional interaction: When a user query is formulated precisely, the system immediately identifies the information required and provides structured project proposals, eliminating the need for further conversation. (b) Conversational, consultative interaction: When faced with vague or incomplete initial prompts, the system employs a functional dialogue strategy. Through targeted clarifying questions, it collaboratively structures the problem space (e.g. project management) to ensure subsequent recommendations are tailored and trustworthy.

This comparison illustrates the system's hybrid role as a de-anthropomorphized tool that balances task efficiency with consultative guidance as the situation demands.

### 5.3 Requirement Cluster 3: Personalization and Neutrality

The requirement for personalization and neutrality in an LLM-based recommendation system reveals a complex field of tension regarding user expectations. A key finding from the interviews was that the nature of the interaction played a decisive role in fostering trust in and acceptance of the system.

The interviewees emphasized that the perception of the bot as a social actor depends on whether it presents itself as anthropomorphic or purely functional.

There was a clear divergence of opinion among the participants ( $n = 11$ ): while four interviewees appreciated a personal touch and human-like dialogue in exploratory or ambiguous situations, seven interviewees—particularly those with experience in academic decision-making—explicitly demanded formal, fact-driven communication.

Throughout all interviews, participants emphasized the importance of high controllability, stating that personalization features should be explicitly adjustable, enabling users to activate or deactivate conversational and anthropomorphic elements depending on the context and task. This control mechanism not only protects privacy but also acts as a tool to counteract the so-called ‘filter bubble’, enabling users to seek out new cross-disciplinary ideas outside their usual area of expertise. For privacy-sensitive individuals, neutrality is often demanded as a standard. For this user type, excessive personalization, such as that based on subtle analysis of writing styles, can be perceived threatening. These users prefer the delivery of factual information, where the LLM-based recommender system acts primarily as an efficient filter, providing concrete result lists without using an unexpectedly familiar tone that might irritate the user. This group requires transparency concerning data processing; the system should offer the option of addressing security concerns directly within the dialogue, rather than interrupting the flow of information with repetitive privacy notices. A gamified approach with different user levels could address this issue by providing a secure environment in which users can decide how much information to disclose. By contrast, innovation-oriented individuals view the system as an empowered service provider. These users are willing to share more detailed data in exchange for greater efficiency and customized results. For this group, the priority is functional added value, which is achieved through optional uploads of documents such as CVs or academic transcripts. Innovation-oriented users are highly willing to participate in user experiments and desire a system that actively adapts to their dynamic linguistic style and exploits all technological possibilities. Consequently, the system must offer a middle ground of a high degree of standardized neutrality and data privacy as a baseline for all, supplemented by customization options for those who create significant value through deep personalization in their scientific work. Ultimately, the LLM recommendation system must fulfil the dual role of a reserved tool and a proactive advisor, while retaining the end user’s decision-making authority over the intensity of personalization.

#### **5.4 Synthesis: Situational Trust Calibration**

Taken together, the three requirement clusters reveal a coherent pattern in the formation and regulation of trust in academic LLM-based recommender systems. Rather than emerging from human-like interaction cues alone, trust is based on users’ ability to verify, contextualize and critically evaluate system outputs. The results suggest a model of situational trust calibration in which trust is not elicited through human-like interaction cues, but rather through adaptive de-anthropomorphization and verifiability. Transparency and source traceability establish a baseline of confidence, while adaptive conversational strategies regulate the degree of dialogic guidance according to the specificity of the user’s query. Personalization acts as a controllable modifier rather than a default trust

mechanism. This synthesis indicates that, in academic recommendation contexts, trust is neither static nor universally increased through anthropomorphic design, but is instead dynamically calibrated through interaction design choices that prioritize accountability, user agency and contextual relevance.

## 6 Discussion

### 6.1 Design Implications for the IdeaLize Platform

This challenges the widely held belief in conversational AI research that greater human-like behavior necessarily leads to greater trust. In high-stakes academic environments, trust appears to emerge not from social bonding, but from the system's willingness to acknowledge its limitations. Building on this insight, the results show that trust in academic contexts is not primarily achieved through anthropomorphic system features, but through deliberate interaction design and system transparency. Although previous research suggests that anthropomorphized LLMs can enhance trust (Maeda, 2024), the present findings indicate that this effect is highly context-dependent in higher education and should not be considered a universal design principle. Importantly, the findings do not suggest that conversational elements are inherently problematic. Instead, they suggest that trust in academic recommender systems should be based on transparency rather than affective social bonding. Instead, striking an adaptive balance between functional de-anthropomorphization and situation-dependent dialogic guidance emerges as a central requirement. In terms of the design of the IdeaLize platform, this means that the system must be clearly presented as a technical decision-support tool. Any anthropomorphic cues that could create the impression of a human expert or an authoritative entity must be avoided. This de-anthropomorphization prevents overtrust and supports a realistic assessment of the system's capabilities, particularly in scientific decision-making scenarios involving institutional responsibility. However, the results also show that a completely neutral, purely functional interaction is insufficient for all usage scenarios. Instead, trustworthy design requires an interaction strategy that adapts to the user's requests. For clearly formulated, fact-oriented enquiries, users expect efficient, tool-like presentation of results, such as structured tile views with relevant data. In this case, dialogic depth is perceived as potential cognitive overload rather than added value. Conversely, vague or incomplete enquiries necessitate a more dialogic and advisory interaction style. In these instances, the system acts as a moderator, using targeted follow-up questions to clarify requirements and structure the problem space before providing concrete recommendations. This form of adaptive conversation should not be understood as traditional anthropomorphism, but rather as functional conversational management that is clearly distinct from deception. Using polite or supportive language is permissible, provided it is not accompanied by attributions of human-like understanding. Crucially, the user always retains the final decision-making authority, which is perceived as a trust-building factor, particularly within an academic context. In summary, the design implications show that IdeaLize should not be conceived as either a purely anthropomorphic dialogue partner or a static search tool. Instead, the platform must adopt a hybrid role, acting as a transparent, non-anthropomorphic tool capable of switching to advisory interaction mode when required. This nuanced design significantly contributes

to the realistic calibration of trust, fostering the long-term acceptance of LLM-based recommendation systems within higher education.

## 6.2 Limitations and Methodological Reflection

While this study provides valuable insights into the design of the IdeaLize platform, several limitations should be borne in mind when interpreting the findings. These limitations primarily concern the characteristics of the sample, the rapid evolution of the technology and the methodological framework chosen.

**Sample and Representativeness.** A significant limitation relates to the composition of the sample. While various stakeholders from the higher education sector were included, the group is small and mainly based in the Heilbronn ecosystem. Perspectives may vary significantly across academic disciplines due to differing departmental cultures and varying levels of technical affinity. Furthermore, there is a risk of self-selection bias, as individuals with an existing interest in AI or digital innovation may have been more motivated to participate. Consequently, the views of AI sceptics or individuals with low technical literacy may be underrepresented.

**Temporal Context of the Findings.** Research in the field of large language models (LLMs) is characterized by extremely short innovation cycles. Participants' perceptions of trust and anthropomorphism are heavily influenced by the current state of technology (e.g. GPT-4). As these systems evolve rapidly and become increasingly integrated into daily academic routines, user expectations are constantly changing. Consequently, these results offer a "snapshot" of a transitional phase in the adoption of academic AI.

**Methodological reflection.** This study primarily relies on interviews. While the qualitative design allowed for deep insights into the 'why' behind user preferences, it does not permit statistical generalization to the entire population of university members. Furthermore, as the interaction scenarios were simulated, a longitudinal field study would be required to determine whether the identified trust factors remain stable during productive use in the real world over several months. The complexity of the trust construct. Trust is a multidimensional phenomenon influenced by more than just interface design (i.e. anthropomorphism versus de-anthropomorphization). Individual psychological traits, prior technological experience and organizational frameworks (e.g. university data protection policies) also play a crucial role. While these external factors were considered, they could not be fully controlled within the scope of this study. Nevertheless, they are likely to have a significant impact on how individual users calibrate their trust in the IdeaLize platform. In summary, these limitations do not diminish the value of the study; rather, they define the framework for future research. Subsequent studies should aim to validate the trends identified here using quantitative longitudinal designs and cross-cultural comparisons.

## 7 Future Research Directions

Rather than asking how human trust can be transferred to LLMs, this work redefines trust as a design issue, focusing on how systems can be created that deserve critical analysis rather than social confidence. From this perspective, future research should move beyond

anthropomorphic metaphors and focus on the conditions that enable LLM-based systems to be used reflectively and accountably.

First, future studies could examine how trust in such systems evolves over time and across repeated interactions. While this study focuses on initial perceptions of trust and design expectations, a longitudinal approach could offer deeper insights into how users adapt their reliance strategies as they become more familiar with LLM-based systems. Second, Secondly, more research is needed to understand how specific system configurations encourage critical evaluation rather than the automatic acceptance of recommendations. Prior work suggests that transparency and explanations alone are insufficient. Therefore, future studies could explore the design conditions under which users are more likely to engage in analytical scrutiny instead of intuitive trust, such as varying degrees of source traceability, uncertainty signaling, or interaction speed. Third, the role of anthropomorphism warrants further investigation. Rather than treating it as a fixed design decision, future research may conceptualize anthropomorphism as a situational and task-dependent interaction strategy. In particular, the question of how adaptive interaction styles can support trust calibration without encouraging overreliance remains unanswered. Finally, future research could examine how organizational and institutional contexts shape trust in LLM-based systems. Differences in governance structures, data policies and disciplinary cultures may significantly influence how transparency, personalization and system framing are perceived and valued by users.

Taken together, these directions outline a research agenda that conceptualizes trust as a dynamic, designable relationship, rather than a static outcome or emotional bond. This relationship is grounded in verifiability, accountability and human oversight.

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